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Posture, muscle activity and muscle fatigue in prolonged VDT work at different screen height settings

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With the increasing use of video display terminals (VDTs), there is growing concern over the corresponding increase in the number of health problems reported. Although much research has focused on identifying the optimal screen height, there is to date no consensus. This study aimed to investigate the effect of prolonged (89 min) VDT work at four different screen heights on head–neck posture, muscle activity and the development of muscle fatigue. The results show that lowering screen height, starting from 15 cm above the baseline (i.e. top of the screen level with eye height while sitting), decreased the ear–eye angle, increased the viewing angle, increased the viewing angle relative to the ear–eye line, and increased the muscle activity of the neck extensor muscles. There were also some significant time effects on postural angles and muscle activity. In this study there were only rare occurrences of muscle fatigue, defined as a simultaneous increase in EMG amplitude and a shift of the EMG power spectrum to lower frequencies. Muscle activity increased significantly in some muscles and for certain screen heights.

1. Introduction

The use of visual display terminals (VDT) has become an activity in daily work for an increasing number of individuals. Although this implies a decrease in traditional forms of manual labour and the associated discomforts, new health problems arise. Musculoskeletal discomfort, for example, has been found to occur and increase during a working day among VDT workers (Hagberg and Sundelin 1986). Another health concern is the visual discomfort while working with a computer screen (Bergqvist *et al.* 1995). Much research has focused on identifying the ideal VDT screen height considering musculoskeletal problems of the shoulder and neck region and visual strain. The requirements to decrease musculoskeletal strain while also decreasing visual strain may conflict, and the feasibility of this remains debatable.

Several authors have studied the effect of screen height and document position on musculoskeletal strain. Hamilton (1996) found that the lowest levels of tension in the neck muscles were experienced with the smallest deviations in head position. Villanueva *et al.* (1997) found a strong significant correlation between posture changes caused by different screen heights and muscle activities. Decrease in neck

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flexion caused a decrease in neck muscle activity ($r = -0.79$ to $r = -0.96$). Turville *et al.* (1998) found that lowering the VDT to increase the viewing angle from 15° to 40° affected the cervical erector spinae muscle significantly, and resulted in higher activity levels. Sommerich *et al.* (2001) also reported an increase in muscle activity with a viewing angle of 35° for different muscles, although it appeared that monitor height had the most influence on the cervical erector spinae muscles. Horikowa (2001) measured an increase in trapezius muscle hardness, but several electromyographic (EMG) data failed to show an increase in trapezius muscle activity, probably due to the support of the forearms (Laursen and Jensen 2000). Trapezius muscle load with forearm support was reported to be below 1% maximal voluntary contraction (MVC) (Aarås *et al.* 1997, Aarås and Ro 1997). Many VDT users report a feeling of fatigue when working at their screens for several hours (Hagberg and Sundelin 1986). In general, these authors agree that a higher monitor placement is required to reduce musculoskeletal strain. Most studies describe a change in muscle activity, but research into muscle fatigue is less common. A longer duration is required to produce and identify muscle fatigue, while most authors have only used a 5–30 min computer task. Turville *et al.* (1998) used a 2 h reading task, but could not detect a significant shift in median frequency (MDF) in the EMG signal, so no fatigue was identified for any of the screen heights studied. An exact screen height, which reduces muscle fatigue, has not yet been established. Further research is therefore required to determine whether the recommendation for musculoskeletal problems and low muscle activity can also be applied to reduce muscle fatigue.

Recommendations for prevention of visual fatigue, however, confuse the issue. The resting point of the eyes is considered to be 15° below the horizontal when the head is upright, so to reduce visual strain a VDT screen placement lower than the horizontal line of sight is recommended. Several authors have described the effects of a downward gaze and a downward viewing angle on eye comfort. Sotoyama *et al.* (1997) reported a higher tear volume while gazing 30° downwards than in the 0 and 15° positions. The proposed cause was a decrease in ocular surface. Larger ocular surface has been found to increase visual strain due to increased tear evaporation. Other variables that affect eye discomfort are vertical gaze inclination and viewing distance (Lie *et al.* 1997). Villanueva *et al.* (1996) reported a correlation between neck angle and screen height and suggested that the neck seems to be the most important component in adjusting the position of the head to find the preferred viewing angle. In their study, significant changes of viewing angle and vertical gaze direction were found for different screen heights. Lower monitor settings gave larger viewing angles and larger vertical gaze directions. Generally it was found that high monitor placement caused more visual strain than lower monitor placement (Villanueva *et al.* 1996, Saito and Sotoyama *et al.* 1997, Sotoyama *et al.* 1997, Burgess-Limerick *et al.* 1998, Psihogios *et al.* 2001). Jaschinski *et al.* (1998) reported that high monitor placement could provoke eye discomfort, especially when the monitor was placed closer to the eyes than the subject's preferred distance. However, viewing distance and gaze inclination did not change significantly with the time of day (Jaschinski *et al.* 1998). Jaschinski *et al.* (1997) reported that during close vision, the ocular muscles are loaded to a lesser extent when vertical gaze is directed downwards. Ankrum (1997) also reported that screen-tilt influences eye discomfort. This study showed that a monitor with the top tipped forward caused more postural and visual discomfort. Taking all factors into consideration these studies seem to recommend a VDT screen height lower than the horizontal. Postural changes with

screen heights during a certain time period have been studied less often. Turville *et al.* (1998) did find a significant increase in postural shifts in a 2 h computer task, but change in head tilt was not examined in their study. More research is needed on this issue to find how working time influences changes in head and body posture.

The effect of different screen sizes on posture was studied by Villenueva *et al.* (1996). They suggested that the use of larger computer screens might induce an upward gaze and a more erect posture. Burgess-Limerick *et al.* (1998) questioned whether the results for normal sized monitors found in their study would hold for different monitor sizes. Sommerich *et al.* (2001) reported lower activity in all muscles using a larger monitor, though for some muscles there was a restriction on this effect, for which no clear explanation was apparent. Previous research has aimed to examine the effects of changing screen height, but in several studies muscle fatigue was not detected. Most authors compared means and did not look for time effects within one specific screen height setting.

Considering the findings described on musculoskeletal and visual strain, the study presented here was designed to compare signs of fatigue in both musculoskeletal and visual aspects for different screen height settings. The main aim was to determine the onset and degree of muscle fatigue within a certain time period while working at a specific screen height, and then to compare these for different screen heights. This same method was also used to examine time effects in postural and visual changes.

2. Method

2.1. *Subject details*

Sixteen subjects (eight male and eight female; 14 right-handed and two left-handed) gave their informed consent to participate in this study. No exclusion criteria were used. The subjects were asked to complete a form providing information of possible relevance to the study results. Seven subjects had a history of musculoskeletal neck and shoulder symptoms during VDT work, and the remaining nine reported no problems. Glasses or contact lenses were used by 10 of the subjects, none of whom used bifocals. The subjects formed a rather heterogeneous group, for whom the demographic data are presented in table 1.

2.2. *Materials*

The VDT workplace was fully adaptable to each individual subject, comprising a table with adjustable height, an adjustable office chair, and a monitor arm. A 17-inch cathode ray tube (CRT) screen was set up at three screen heights, and a 15-inch Thin Film Transistor (TFT) flat panel display for a laptop computer was used for the lowest placement. A keyboard and mouse were used at all screen heights. EMG data were gathered with a telemetric EMG-device (model ME3000P8, Mega Electronics Ltd., Kuopia, Finland). The pre-amplified EMG signal (analogue differential amplifier) was measured at a sample rate of 1000 Hz. The signal bandwidths of the high- and low pass filters were 8 and 500 Hz respectively. Posture was analysed with a Sony Hi 8 video camera, which digitized the data to measure several angles between reflective markers placed on the subjects.

2.3. *Task*

All subjects worked for about 89 min at each of the four different VDT screen heights. The test sessions were spread over 2 days due to the length of each session. Subjects had to work at two screen heights in 1 day and were asked to return 1 week

Table 1. Demographic subject information and experimental workstation measurements.

Variables	Mean ± SD	Minimum	Maximum
Age (years)	24.0 ± 4.7	19	39
Height (cm)	176.6 ± 10.5	160.0	192.0
Weight (kg)	68.9 ± 9.2	54.0	82.0
Chair height from floor surface (cm)	49.9 ± 2.9	45.0	54.5
Table height from floor surface (cm)	70.4 ± 4.0	65.0	79.0
Keyboard distance from the edge of the table (cm)	12.4 ± 3.8	6.5	18.0
VDT – Baseline screen height above table surface (cm)	54.7 ± 3.5	50.0	63.0
Mean duration of VDT work per day in normal work conditions (h)	3.6 ± 2.7	0.5	8.0

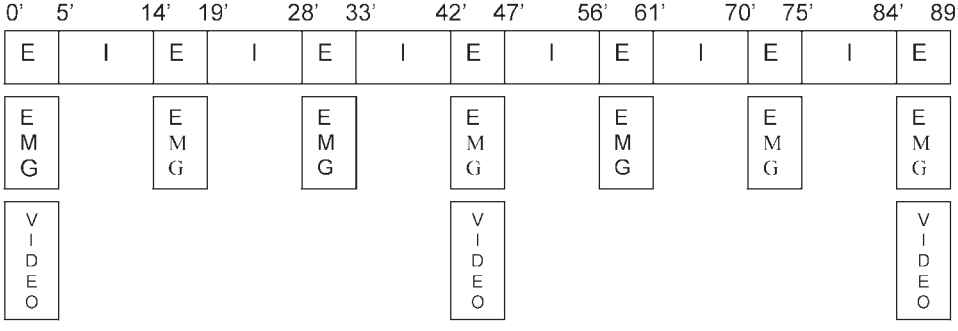


Figure 1. Schematic representation of experimental protocol.

later, on the same day and at the same time. Between screen heights, subjects rested for at least 45 min. To neutralize the effect of the sequence of the different test situations, the order of the change in screen height was completely randomized. The work task included an experimental part (E) and an Internet based search task (I). A schematic representation of the experimental protocol is shown in figure 1. During the experimental part of the task, the subjects played a standardized keyboard computer game for 5 min. The game required visual monitoring and control of the arrow keys on the keyboard, using the right hand only. Subjects rested their left hand on the table. During this part, the raw EMG signal and video images were recorded and subjects were asked to make no abrupt movements. The Internet based search task lasted 9 min and subjects were allowed to reposition themselves in the chair and to move neck and shoulders if they preferred to do so. During this part of the experiment, the subjects used the keyboard and computer mouse. The intention was to create a situation similar to their normal VDT working practices.

2.4. *Experimental procedure*

Following some reference voluntary contractions (RVCs), the subject sat at an adjustable workstation. Firstly, the workplace (office chair and table height) was adjusted to the subject's personal anthropometrics. Seat base height was adjusted

so that the subject's thighs were horizontal, lower legs vertical, and feet flat on the floor. The height of the armrest was adjusted and permitted an elbow angle of 90°. The keyboard was situated at elbow height, so that with upper arms vertical, elbow bent 90°, and fingers comfortably extended, the subject's fingertips reached the numeric row (see figure 2). The VDT screen was then raised until the top of the screen was exactly level with the subject's eyes. This height is referred to as the baseline screen height, and was measured from the table surface to the top of the screen. For the high placement, the screen was positioned 15 cm above the baseline, and for the low placement the screen was positioned 15 cm below. This adapted the conditions to the individual and the VDT screen could be positioned in a circle around each subject.

Straker *et al.* (1997) found that subjects using laptop computers instead of desktop computers had increased neck flexion and head tilt. They also reported a clear trend, although not significant ($p = 0.08$), for greater discomfort after laptop use than when using a desktop computer. Villanueva *et al.* (1996) showed that the use of laptop computers placed on the table surface caused not only more musculoskeletal, but also visual discomfort and increased neck extensor muscle activity. Because of these findings the height of a laptop monitor placed on the table surface (laptop position) was included in this study to allow comparison between the three desktop monitor heights and a flat panel display, as often used in office situations. The height of the top of the laptop screen above the table surface was about 27 cm. The laptop was placed behind the



Figure 2. Subject seated at test workstation with the highest screen position.

keyboard required for the task. Horizontal distance between the eyes and the middle of the screen was standardized at 60 cm, except for the laptop screen height, which varied because of differences in subjects' height. All 16 subjects performed an 89-min task at each of the previously described screen heights. Three types of data were collected: electromyography, video images and subjective scores of fatigue.

2.4.1. *Electromyographic analyses*: The raw EMG signal was recorded with a portable EMG device (model ME3000P8, Mega Electronics Ltd., Kuopia, Finland) for 5 min, seven times within the 89 min of each session. Bipolar Silver-Silver Chloride surface electrodes (Nikomed, Denmark) were placed on eight muscles (see table 2) situated around the neck – shoulder region. The inter-electrode distance was 20 mm and placement of the electrodes conformed to Seniam recommendations (Hermens *et al*. 1999) and recommendations by De Luca (1997).

Two EMG parameters were extracted from the raw EMG signal. The averaged rectified EMG (AEMG) was calculated (time window = 0.5 s) to quantify the muscle activity. Myoelectrical signs of muscle fatigue are characterized by a simultaneous increase in AEMG and decrease in MDF (Luttmann *et al*. 2000). Therefore, the median frequency (MDF) was calculated using a Fast Fourier Transformation (FFT). The slopes (% change per time unit) of the EMG parameters, determined using linear regression techniques, quantified the EMG changes over time.

Comparison between subjects and test sessions was possible by normalizing all EMG parameters as a percentage of the RVC. The RVC activity for the trapezius muscle (TRP) was a 90° abduction of the arm with a 1 kg weight in the hand of the arm being tested. The RVC activity for the deltoid muscle (DLT) was a 90° flexion of the arm in the sagittal plane with a 1 kg weight in the hand of the arm being tested. The RVCs for the sternocleidomastoïdeus muscles (SCM) and the neck extensor muscles (NXT, splenius capitus) were tested with the subjects lying on a treatment couch in supine and prone positions respectively, with the neck and head extending beyond the end of the couch, and therefore supported by the neck muscles alone. In this way the weight of the head acted as an external load for the neck muscles. All RVC tests were taken before the sessions, and measurements were recorded for 5 s for each test. The RVC tests for the TRP and DLT muscles were separate, so that each arm could be tested separately, while the SCM and NXT muscles were tested in one 5-s test session.

Table 2. EMG measured muscles.

Muscle	Code
Right m. trapezius pars descendens	TRPr
Right m. deltoïdeus pars anterior	DLTr
Right m. splenius capitis	NXTr
Right m. sternocleideomastoïdeus	SCMr
Left m. trapezius pars descendens	TRPl
Left m. deltoïdeus pars anterior	DLTl
Left m. splenius capitis	NXTl
Left m. sternocleideomastoïdeus	SCMl

2.4.2. *Posture analyses with video images*: During EMG measurement, posture and workstation settings were captured by a video camera (Sony Hi 8 video camera). These images were taken on the first, fourth and seventh 5-minute EMG measurement period and digitized on the 30th, and 150th and 270th second within the 5-minute period (300 s), a method used previously by Turville *et al.* (1998). Differences between screen heights were tested by taking the data from the 4th EMG measurement on the 45th minute of each session.

The following angles were analysed (see figure 3):

- the Ear – Eye Angle (EEA), between the horizontal and a line joining the outer cantus of the eye and the trages of the ear;
- the Viewing Angle (VA), between the horizontal and a line joining the outer cantus of the eye and the middle of the computer screen;
- the Viewing Angle relative to the Ear – Eye line (VArel).

Angles below the horizontal plane are indicated with a minus sign.

Repeated digitizing of the same captured video image, revealed variations in VA, EEA and VArel of about 0.4°, 3° and 3° respectively depending on the distance between the markers. The use of more precise instruments might lead to more accurate measurements. The results presented in 3.2 show that the method was sufficiently accurate to detect the influence of VDT screen height on the calculated angles.

2.4.3. *Measurements of subjective feelings of discomfort*: To give an indication of the subjective feelings of discomfort of the eyes, the neck and both shoulders, the Borg CR-10 rating scale (Borg 1982) was used. This scale has ratio properties using numbers and verbal expressions. The Borg scale has been used by several authors to describe both discomfort and fatigue (Hagberg and Sundelin 1986, Sundelin and Hagberg 1992, Hermans and Spaepen 1995, Swanson *et al.* 1997, Liao and Drury

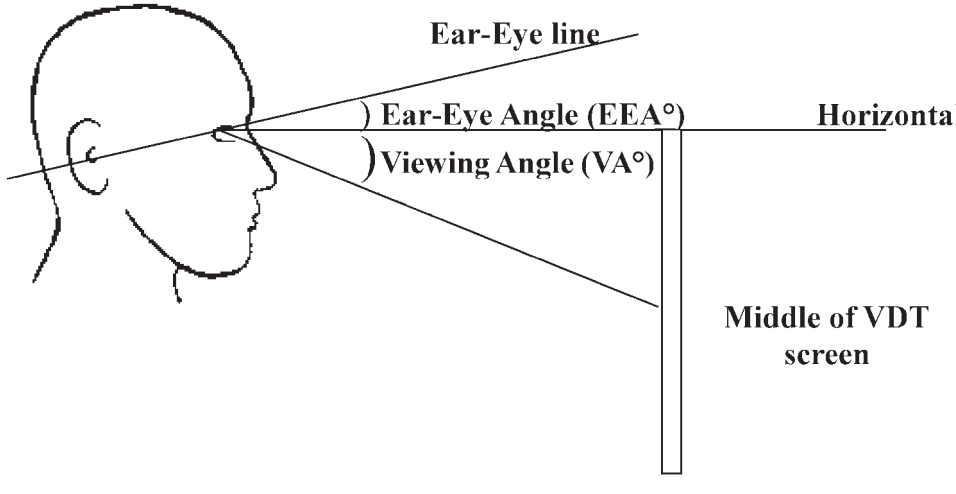


Figure 3. Postural and viewing angles relative to the horizontal shown in the baseline workstation arrangement (top of the screen level with eye height while sitting).

2000, Psihogios *et al*. 2001). Immediately before starting the session, the subjects were asked to rate their subjective feelings of discomfort in the eyes, the neck, and the shoulders. During the experiment this rating was repeated every 5 min to see whether discomfort changed with time and to evaluate whether the end score varied for the different screen heights.

2.4.4. *Statistical analyses*: Pearson correlation and Student's *t*-tests were used to examine the relationship between the different anthropometric and medical data of the sample. EMG data were analysed using One-Way ANOVA with a Repeated Measures design. If significant, a Tukey HSD *post hoc* test was used to determine specific differences between test heights. In addition to ANOVA, a Linear Regression Technique was used to detect time effects. The postural changes were also analysed using One-Way Repeated Measures ANOVA. The differences in Borg scale ratings were tested using a Friedman ANOVA test for non-parametric data. The level of significance was chosen as $p \leqslant 0.05$ for all statistical tests. Unless otherwise stated, all data are expressed as the mean $\pm$ SD.

3. Results

3.1. *Subjects*

Study of the subjects showed that the group with a history of shoulder or neck problems had worked for a significantly longer period behind VDT screens than the group with no complaints ($p < 0.01$). In this sample there was an obvious correlation between age and time spent behind the computer per day ($r = 0.69$, $p < 0.01$), probably because the subjects above 24 years of age had computer-task related jobs.

3.2. *Posture*

Table 3 shows the mean $\pm$ SD values of postural angles with the different screen heights. Change in screen height significantly affected Viewing Angle (VA), Ear–Eye Angle (EEA) and Viewing Angle relative to the Ear–Eye line (VArel). In particular, the VA decreased when the VDT screen was in a higher position ($p < 0.01$) and the eyes needed to be directed more upwards. There was a greater EEA with higher screen heights ($p < 0.01$), so the head was more erect. The VArel also showed significant differences between the different heights, being larger for lower monitor height settings ($p < 0.01$).

Table 3. Mean $\pm$ SD values of postural angles at different screen heights after 45 min of VDT work

Screen Height	Viewing Angle (VA°)	Ear–Eye Angle (EEA°)	Viewing Angle relative to Ear–Eye Line (VArel°)
BASELINE + 15 cm	3.91 $\pm$ 1.18	22.53 $\pm$ 7.03	18.61 $\pm$ 6.45
BASELINE	− 7.56 $\pm$ 1.66	14.64 $\pm$ 5.99	22.19 $\pm$ 5.93
BASELINE − 15 cm	− 16.54 $\pm$ 1.63	9.07 $\pm$ 8.64	25.61 $\pm$ 7.83
LAPTOP	− 23.50 $\pm$ 2.00	3.98 $\pm$ 8.16	27.48 $\pm$ 7.20
One-Way ANOVA Significance level	$p < 0.01$	$p < 0.01$	$p < 0.01$

Note: Angles below the horizontal are indicated with a minus sign.

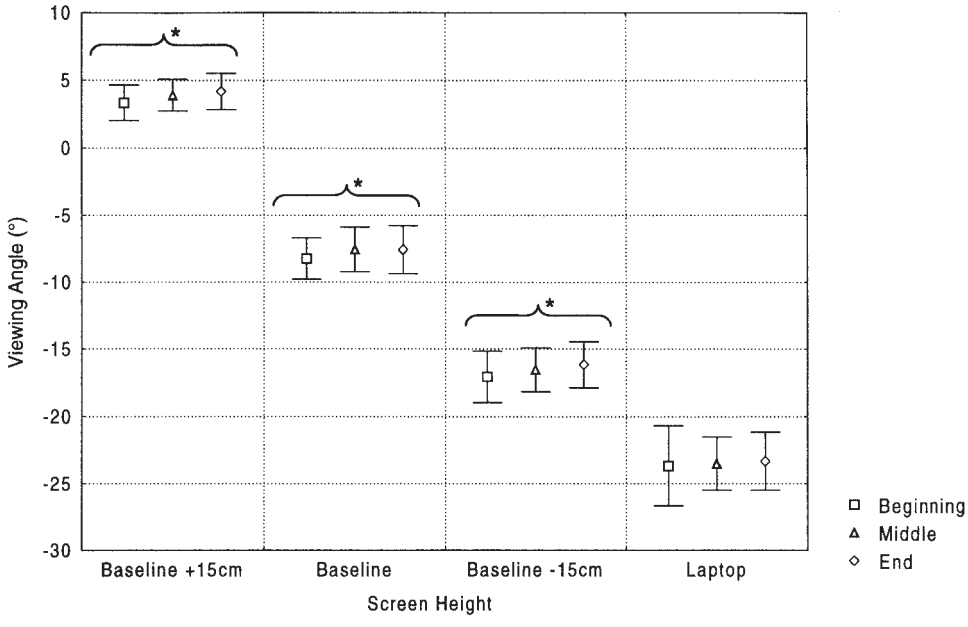


Figure 4. Mean ± SD viewing angle (VA°) after 5 minutes (Beginning), 45 minutes (Middle) and 90 minutes (End) of VDT work at different screen heights (*$p < 0.01$).

In addition to differences in heights influencing the angles, time also had an influence. Figure 4 shows the VA (mean ± SD) at the beginning, middle and end of the test session for the different screen heights. During the 89-min test period the VA increased significantly ($p < 0.01$), except for the laptop condition. The EEA only increased significantly from (mean ± SD) 21° ± 6 to 24° ± 6 above the horizontal plane in the high VDT screen condition ($p \leqslant 0.05$). The VArel did not change significantly with time.

3.3. *Muscle load at the different screen heights*

Table 4 presents the mean normalized muscle activities at the different screen heights during the 4th EMG measurement. Muscle activity only changed significantly in the right and left NXT muscles ($p < 0.01$). The activities of the other registered muscles were rather low and not affected by screen height. Muscle activity differed significantly between the right and left trapezius ($p \leqslant 0.05$). Table 4 shows that the right trapezius muscle was consistently more activated than the left, at all screen heights.

3.4. *EMG changes over time*

In this study there were only rare occurrences of muscle fatigue, defined as a simultaneous increase in EMG amplitude and a shift of the EMG power spectrum to lower frequencies. On the other hand, an increase in muscle activity, quantified by the temporal changes in AEMG, during the 89-min test session occurred for some muscles and screen heights. Table 5 shows the muscles and the corresponding screen heights in which the relative slope of the AEMG parameter was significantly ($p \leqslant 0.05$) or almost significantly ($p \leqslant 0.10$) different from zero. Prolonged work at

Table 4. Mean ± SD normalized muscle activity (% RVC) of the registered muscles at different screen heights during the fourth EMG measurement (after 45 min).

ScreenHeight	TRPr	TRPl	DLTr	DLTl	NXTr	NXTl	SCMr	SCMl
Baseline + 15cm	18.03 ± 23.09	10.54 ± 11.85	1.45 ± 1.17	1.58 ± 0.97	29.35 ± 14.17	26.14 ± 11.19	3.92 ± 2.43	3.63 ± 1.81
Baseline	13.98 ± 10.35	8.29 ± 6.19	1.52 ± 1.22	1.42 ± 0.67	31.47 ± 14.6	29.28 ± 11.6	4.12 ± 3.23	3.16 ± 2.87
Baseline – 15cm	19.76 ± 21.25	13.22 ± 14.80	1.19 ± 0.83	1.94 ± 1.25	37.95 ± 14.41	37.24 ± 15.41[ab]	3.97 ± 1.9	3.76 ± 2.03
Laptop	13.26 ± 11.03	7.69 ± 7.38	1.88 ± 1.65	2.19 ± 2.4	44.47 ± 11.02[ab]	38.06 ± 9.53[ab]	4.93 ± 3.99	4.25 ± 3.11
One-Way ANOVA Significance level	NS	NS	NS	NS	$p < 0.01$	$p < 0.01$	NS	NS

[a]Significantly different ($p < 0.05$) compared to the Baseline + 15 cm condition. [b]Significantly different ($p < 0.05$) compared to the Baseline condition

Table 5. Muscles and screen heights for which the relative slope of the AEMG parameter differed significantly ($p \leqslant 0.05$) or almost significantly ($p \leqslant 0.10$) from zero.

Muscle	Screen height	Mean % change per measurement period (min – max)	p-value (linear regression analysis)
TRPr	BASELINE + 15 cm	0.92 (–1.7 – 5.57)	$p = 0.07$
	BASELINE	0.57 (–1.84 – 2.16)	$p = 0.09$
	BASELINE – 15 cm	0.85 (–1.11 – 2.16)	$p = 0.10$
	LAPTOP	0.54 (–0.94 – 3.86)	$p = 0.05$
NXTr	BASELINE + 15 cm	0.57 (–1.39 – 2.4)	$p = 0.04$
	BASELINE	0.50 (–2.89 – 2.16)	$p = 0.09$
	LAPTOP	0.40 (–1.96 – 3.57)	$p = 0.09$
SCMr	BASELINE + 15 cm	0.11 (–0.011 – 0.50)	$p = 0.03$
	BASELINE – 15 cm	0.08 (–0.28 – 0.55)	$p = 0.02$
NXTl	BASELINE + 15 cm	0.48 (–2.01 – 9.82)	$p = 0.07$

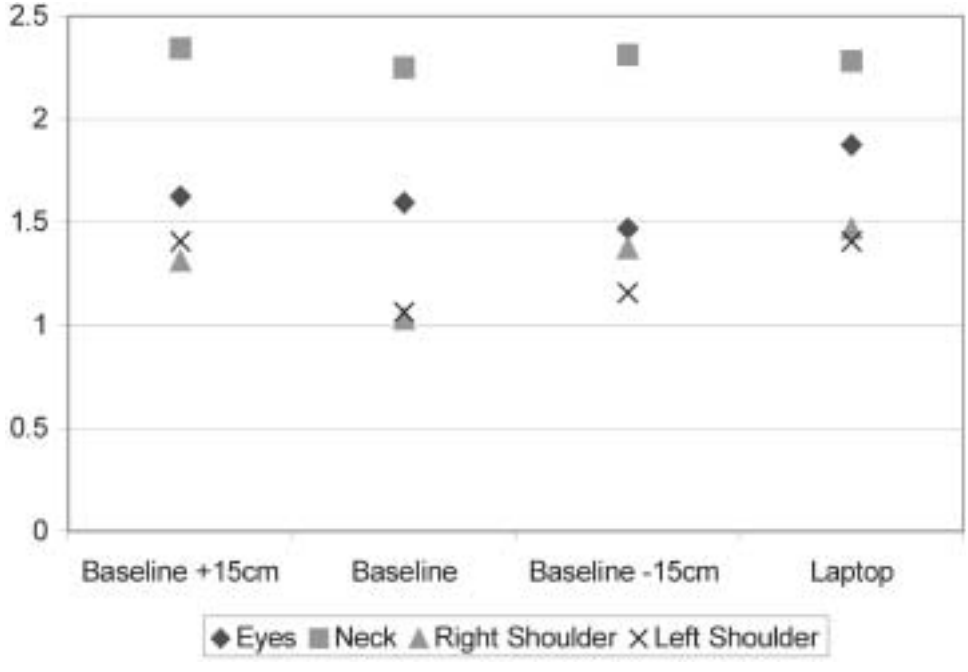


Figure 5. Mean end scores (after 89 min VDT work) from the Borg rating scale for discomfort in the eyes, neck and shoulders with different screen heights.

the highest screen height (baseline + 15 cm) seems to have the most significant effect on the increase in muscle activity.

3.5. *Subjective rating of discomfort*

The end scores on the Borg rating scale (Borg 1982) for discomfort in the eyes and neck – shoulder region varied between 0.5 (very, very weak) and 5 (strong). In all four tests, the subjects reported the most discomfort in the neck region (see figure 5). However, these were only trends with no statistically significant differences between screen heights.

4. Discussion

The heterogeneous sample in this study could make external validity rather high, taking into account the fact that the test group was small ($n = 16$) and mean age was 24 years.

4.1. *Posture*

The observed mean postural angles were similar to those reported in previous studies. The EEA determined by Jampel and Shi (1992) was 15° above the horizontal line of the eyes. Grey *et al.* (1966) found an EEA of 14.5° in erect posture. In the present study a similar 14.6° EEA was determined. The results show a significant change in posture when screen height was altered. The VA, EEA and VArel differed significantly between all screen height settings. VA and VArel increased with lower monitor positions while EEA decreased with the VDT screen in the low placement, confirming previous research (Villanueva *et al.* 1996, 1997, Burgess-Limerick *et al.* 1998, 2000, Psihogios *et al.* 2001, Sommerich *et al.* 2001). Time affected posture in VA and EEA (Table 3). VA significantly decreased with time in the high, baseline and low VDT screen positions. There was no difference in the laptop condition. EEA increased only in the high monitor position (21° – 24°). The time effects in VA could not be attributed to a change in head tilt in the sagittal plane, because EEA did not change significantly with time for the other screen heights. Therefore, the change in VA was probably due to subjects slouching in their chairs during the 89-min test sessions. This slouching would cause the eyes to be positioned at a lower level, relative to the VDT screen, so VA appeared to decrease. In the highest screen condition the EEA did change significantly with time because the eyes could not compensate in the high screen placement. This possibility is logical, as upward gaze is not recommended (Sotoyama *et al.* 1997).

A fourth angle, which could not be measured in this study, is the gaze angle. Gaze angle is defined by the position of the eyes and can be determined by measuring the ocular surface (Villanueva 1996). However, only part of the changes in VA can be attributed to gaze angle changes, because EEA does not fully account for the changes in VA (Villaneuva *et al.* 1996, Burgess-Limerick *et al.* 1998). Sommerich *et al.* (2001) found a 13° compensation for gaze angle in a total 35° change of VA. In our study a total gaze angle compensation of 9° was found within a 27° change of VA. Although significance between screen heights was not tested, the highest contribution (38%) of gaze angle to the change in VA was found between baseline and low VDT screen heights. Between the low VDT screen height and laptop monitor height the contribution of the gaze angle was only 27% while between the highest and baseline position 31% was found. This means more flexion of the neck was required, and less change in gaze angle was used below a VA of − 17°. These results for low monitor placement confirm research by Sommerich *et al.* (2001). Since screen heights higher than the baseline were examined in this study, it is concluded that more head tilt and less change in gaze angle (31%) are used when working with VDT screens above baseline, and this is consistent with the resulting time effects on posture described above.

4.2. *Muscular load*

Muscle activity was affected by screen height, but this influence was only significant in the NXT muscles ($p < 0.01$). With the lowering of the computer screen from 15 cm above baseline to table height, there was a 50% increase in NXT muscle

activity, mainly caused by the increase in neck flexion. Already in the condition with the lowest EMG activity of the NXT muscle, the subjects used 25–30% of the amount of muscle activity required to hold the weight of the head (in the RVC test described in 2.4.1). *Post hoc* tests identified which heights showed significant differences (see table 4). The highest and the baseline screen heights showed a significantly lower activity in the right NXT muscle compared to the laptop condition ($p < 0.01$). The left NXT muscle showed significantly lower muscle activity between the highest screen height and the low and the laptop conditions ($p < 0.01$), and in the baseline screen height compared to the low and the laptop conditions ($p \leqslant 0.05$). Comparable effects were reported by Hamilton (1996), Villanueva *et al.* (1997), Turville *et al.* (1998) and Sommerich *et al.* (2001). The differences with the three VDT screen heights and the laptop screen height were also comparable to results on NXT muscle activity found by Villanueva *et al.* (1998). However, it was not the aim of the present study to make a comparison between CRT and flat panel displays, but to determine how muscle activity was affected when a laptop was placed on the table surface, as often practiced in many work environments. In conclusion, the highest level of NXT muscle activity occurred in the laptop condition, particularly caused by the extreme flexion in the neck.

There was much variation in the activity of the other muscles (see table 4), so that a clear trend could not be identified, possibly explained by the low activity levels that were measured. In particular the very low activity in the anterior deltoid muscle is remarkable but understandable, since it is the muscle that is expected to be the least affected by changes in screen height. This muscle was included in the study because it has rarely been investigated elsewhere, and it was of interest to know whether prolonged use of the keyboard and mouse would affect its activity level. However, deltoid muscle activity was hardly affected in this study. Another reason for the low levels of muscle activity could be the fact that the workstation was customised to the subjects to ensure good support and comfort. In this study, the adjustment of the arm rests of the office chair and the set up of the table height provided a full support for the forearm (see figure 2). This argument is supported by Aarås *et al.* (1997) who reported that trapezius muscle activity was significantly lower when forearm support was provided. Laursen and Jensen (2000) also reported that the lack of significance in their study was probably due to the low trapezius muscle activity caused by forearm support. Nevertheless, a significantly higher level of trapezius muscle activity was detected in the right trapezius muscle compared to the left muscle ($p \leqslant 0.05$), probably attributable to the right-handed control of the cursor keys during the experimental part of the study.

4.3. *EMG changes over time*

The myoelectrical signs of muscle fatigue are characterized by a simultaneous increase in AEMG and decrease in MDF (De Luca 1997, Clarys 2000, Luttmann *et al.* 2000). These indications of muscle fatigue were hardly apparent for any of the muscles and screen heights and can probably only be detected with longer test periods. Turville *et al.* (1998) even suggested recording the EMG signal during a whole working day. Nevertheless, a significant increase in muscle activity over the course of the test session, quantified by the slope in AEMG, could be detected for some muscles and screen heights. However, an increase in EMG amplitude can be caused by several factors including an increase in the force production of the muscle, postural changes or muscle fatigue (Luttmann *et al.* 2000). Muscle fatigue based on

an increase of the EMG amplitude is only possible when the force production during the task is kept constant in combination with minor changes in working posture. In this study EMG parameters were extracted from the measurements where the subjects played a standardized computer game, to ensure a comparable workload of the task during each of the seven measurement periods. Additionally, the time-dependent changes in the position of the head were minimal. This is reflected in the absence of time effects in the EEA, except for the highest VDT placement condition. For this condition, it is doubtful whether the increase in AEMG can be seen as an indication of muscle fatigue. Table 5 shows the muscles for which the electrical activity (AEMG) changed with time. The largest increase in muscle activity was in the highest monitor position (baseline + 15 cm). At this screen height, AEMG showed a significant ($p \leqslant 0.05$) or almost significant ($p \leqslant 0.10$) increase during the 89-minute test session for four of the eight recorded muscles (right TRP, right NXT, left NXT and right SCM). Irrespective of the screen height, the most remarkable increase in electrical activity occurred in the TRPr muscle. After 89 min of VDT work, the muscle activity of the TRPr muscle increased by about 5%. If this increase continued linearly over an 8-h working day, a quarter more muscle activity would be required to perform the same task at the end of the day. This finding reinforces the work of Hermans and Spaepen (1995), who also found an increased EMG activity of the trapezius muscle at the end of the working day. Although no postural changes of the shoulder were analysed in the present study, elevating the right shoulder while manipulating the cursor keys with the right hand during the computer game may have caused the detected increase over time.

The simultaneous increase in muscle activity of the right NXT and SCM muscle is also noteworthy. The increase in both muscles might be explained by the increase in EEA with time. Adjustments of the office chair prevented subjects from being able to lean backwards. The centre of gravity of the head may therefore have been positioned in front of the rotation axis. Based on this possibility, it could be assumed that neck extension was caused by a contraction of right and left NXT muscles (see table 4). The increase in muscle activity of the right SCM could be due to co-contraction. Therefore, prolonged VDT work at this high monitor position might be harmful.

4.5. *Visual discomfort*

Although not statistically significant, a tendency towards more eyestrain was found in the laptop condition. More eye discomfort being reported while working in the laptop condition could be related to a combination of several factors. Firstly, the size of the laptop screen was somewhat smaller than the CRT monitor (Villanueva *et al.* 1998). However, the visible diagonal size of a CRT monitor is always smaller than the tube's actual diagonal size. A 17-inch CRT monitor has an edge area and the visible diagonal is approximately 16-inches. TFT monitors do not have an edge area, which means that a 15-inch TFT monitor is almost the same as the 17-inch CRT monitor. Secondly, a lower brightness of TFT monitors was noticed but the resolution was kept constant. Additionally, the absence of flickering, geometry and convergence errors due to the technical nature of TFT displays might even improve the visual comfort for the user. Finally, a greater horizontal distance from the eyes to the middle of the screen in the laptop condition (mean $\pm$ SD distance = 76 $\pm$ 5 cm), compared to the 60 cm with the other screen heights, may have contributed to the reports of more eye discomfort.

5. Conclusions

From this study, we can conclude that the height of the VDT screen has an influence on VA, EEA, and VArel and muscle activity of the left and right NXT muscles. In addition, the most remarkable increase in muscle activity over the course of the 89-minute test session was with the highest screen height (baseline + 15 cm). Despite the occurrence of some postural changes with time at this screen height position, the EMG activity increase is interpreted as a sign of muscle fatigue, although further research is needed to clarify this.

Finally, with respect to physical and visual comfort, a VA between 8° and 16° below the horizontal plane is recommended from the results. These recommendations should be generalized with care. Results from this study, using larger VDT screen sizes (> 15 inches), are comparable to previous findings using 14-inch or 15-inch monitors, although less significance was found in the other muscles tested in the study. This can be attributed to the careful arrangement of the workplace according to the subject's individual anthropometry.

In future studies longer test periods, sensitive EMG equipment and postural control may be needed to detect muscle fatigue during VDT work. However, it is not possible to regulate all the confounding factors. From a practical point of view, it is may be better not to disturb people's working practices. It is possible that VDT users even try to alleviate musculoskeletal discomfort and/or prevent fatigue by making non-work related body movements (Karwowski *et al.* 1994). The myoelectric signs of muscle fatigue in low-level, long duration contractions are also still unclear. Individual differences in fiber type distributions and/or force control strategies might make a person more or less sensitive to musculoskeletal disorders.

Muscle activity, muscle fatigue, physical and visual comfort should all be taken into account in recommending a general screen height for a group of VDT workers, and workstations need to be adjusted to the needs of each individual to regulate musculoskeletal and visual discomfort.

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